

# Are we Doomed to an AI Race? Why Self-Interest Could Drive Countries Towards a Conditional Pause on Superintelligence

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September 17, 2026

## Abstract

This paper uses game theory to argue that, contrary to the prevailing view, a conditional pause on Artificial Superintelligence (ASI) can be in a state's self-interest. By formalizing strategic interactions between geopolitical superpowers, we model the trade-off between the benefits of technological supremacy and the catastrophic risks of uncontrolled ASI. The analysis reveals that as the perceived cost of loss of control increases sufficiently relative to other parameters, it becomes in each state's self-interest to impose a pause. We further discuss descriptive indicators, including expert statements, public opinion and institutional developments, that are consistent with rising concern about loss of control, and consider what such a rise would imply for the strategic space for a conditional pause.<sup>1</sup>

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<sup>1</sup>This manuscript is a preprint and has not undergone peer review.

# 1 Introduction

The development of Artificial Superintelligence (ASI), an AI system that greatly exceeds the cognitive performance of humans in virtually all economically relevant domains [4], is considered by leading researchers as a potentially transformative turning point in human history [26, 32].<sup>2</sup> The possible benefits are enormous: breakthroughs in science, medicine, and economic productivity.

At the same time, experts warn of a possible loss of control over ASI [4, 17]. This loss could have multiple sources: specific capabilities such as deception, manipulation, and coercion, or the ability to self-replicate and improve autonomously beyond human intervention. Should such a system operate outside meaningful human control, it may pursue objectives misaligned with human values, taking actions that are deeply undesirable for large portions of humanity, both present and future generations.

This has led to calls for a pause on the development of ASI. In October 2025, the Future of Life Institute (FLI) published the *Statement on Superintelligence*, calling for a prohibition on the development of superintelligence “until there is broad scientific consensus that it will be done safely and controllably, and with strong public buy-in” [10]. This statement was signed, among others, by the so-called ‘godfathers’ of modern AI Geoffrey Hinton and Yoshua Bengio and Apple co-founder Steve Wozniak. According to FLI polling, 64% of Americans believe that superintelligence should not be developed until it is demonstrably safe and controllable [11]. At the level of desirability, the proposal thus has broad support.

Most criticism, however, focuses on infeasibility. Critics argue that, even though states might be better off with a mutual pause, each state individually has the incentive to defect and continue development [3]. That is, it is in a state’s *self-interest* to keep on racing towards ASI and therefore a pause is unfeasible.<sup>3</sup> If the US stops while China continues, it falls behind in a competition for geopolitical supremacy. Racing therefore remains the dominant strategy, even when both parties acknowledge this could lead to

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<sup>2</sup>Note that there is no agreed upon definition of ASI. However, this general definition is the least controversial.

<sup>3</sup>According to the argument, a policy being in a state’s self-interest is necessary for it to be feasible. However, it does not necessarily claim it is sufficient. Among others, practical limitations, such as the lack of verification mechanisms of AI agreements, could also affect the feasibility of a mutual pause on ASI [16].

catastrophic outcomes. The American Enterprise Institute (AEI) illustrated this argument by pointing to Alibaba CEO Eddie Wu, who announced a *roadmap toward artificial superintelligence* with over 53 billion dollars in investments around the time of the Statement. A unilateral pause would therefore, according to AEI, “delay innovation, empower bureaucracies, and hand an enormous advantage to China” [29].

In this paper, building on previous literature about the dynamics of AI development [22, 1, 2, 39, 7], we turn to game theory to examine the assumption that pausing ASI development is contrary to a state’s self-interest. We model a pause as conditional: a pausing state halts frontier development *until* safety conditions are met, after which they resume with maximal safety. By formally modeling the strategic dynamics between geopolitical superpowers, we show that a pause is not necessarily contrary to a state’s self-interest. On the contrary, under certain conditions, the game shifts from a race to develop to a game where it is in a state’s self-interest to introduce a pause on the development of ASI. We assess the possible payoffs for a pause and a racing strategy, based on multiple variables: the capability lead, the size of the winner’s advantage, the decisiveness of capability lead, and the risk and speed advantage of racing. The key variable is the cost of loss of control. When the severity of a potential catastrophic outcome is high enough, that is, when the cost from an uncontrolled ASI exceeds the expected benefits of geopolitical supremacy, strategic space emerges for a pause. While the strategic space for a pause currently appears limited, there are indications that this assessment is beginning to shift. Recent developments in AI capabilities, increasing incidents of unpredictable behavior, and growing consensus within the scientific community about control problems are beginning to change the discourse [11, 27, 13, 25, 37].

We proceed as follows. In Section 2, we justify the game-theoretic approach and set up the model. Section 3 identifies four possible strategic worlds and examines their sensitivity to the model parameters. In Section 4, we analyze descriptive indicators that are consistent with rising concern about loss of control. In Section 5, we address some limitations and possible extensions of the model. Throughout the paper, we keep our discussions informal for accessibility. However, we added an Appendix with all mathematical details.

## 2 A Game-Theoretic Framework for ASI Development

The debate over the feasibility of an Artificial Superintelligence pause is based on a strategic problem involving multiple actors with conflicting incentives. To move beyond speculative debate, it is desirable to set up a formal framework to analyze whether and under which conditions it becomes in the self-interest of states to pause or race. Game theory provides precisely this framework.

Game theory is the mathematical study of strategic decision-making [28]. It works by constructing simplified models of real-world strategic situations, including only the variables relevant to the problem at hand. In our model, states have one shot to choose which strategy to play. Both states know all parameters and choose their policy at the same time, without observing the rival’s choice. They are assumed to act rationally, meaning they play the strategy that best serves their preferences given what they expect the other to do. We assume these preferences to be structured by self-interest, which in the case of states, concerns mainly survival, prosperity, or position within the international system [12]. To solve the game, we analyze pure-strategy Nash equilibria, which are combinations of policies from which no state can gain by unilaterally changing its own. An equilibrium identifies which outcomes are stable given the states’ incentives.<sup>4</sup>

This analytical lens has a well-established track record in global security studies, from modeling arms races and nuclear deterrence [33, 34, 20] to recent applications examining the competitive dynamics of AI development. Foundational work by Armstrong et al. [2] first formalized race to the bottom dynamics for ASI development, illustrating how competitive pressures and information hazards, that is, knowing too much about a rival’s progress, increase the likelihood of an ASI-related catastrophic outcome by players skimping on safety-investment. Katzke and Futerman [22] and Young [39] shifted this paradigm by arguing that the race is a trust dilemma, rather than a prisoner’s dilemma, arguing that since a race maximizes existential threats like great power conflict and loss of control, mutual restraint is the only strategically sound path for rational states. Important challenges to the assumption of the decisive advantage of winning the race were made

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<sup>4</sup>Note that equilibria do not by themselves predict which of several equilibria states will coordinate on, nor do they address verification or enforcement.

by Dung and Hellrigel-Holderbaum [7]. Recently, both strategic paradigms were bridged in one model for AGI development by Abraham et al. [1], who compare state strategies, accounting for interim returns on investment, beliefs about the costs and about AGI timelines.

Our paper makes four contributions relative to this literature. First, we model a policy choice specifically focusing on an ASI pause. Secondly, whereas Armstrong et al. [2] model safety as a continuous investment and treat disaster as weighing less than losing a winner-takes-all race, we model a discrete policy choice between racing and a conditional pause, and allow the cost of loss of control to exceed the value of a safe win ( $C > 1$ ). Thirdly, we model both the speed advantage of racing ( $b$ ) and the risk of racing towards a catastrophic outcome ( $1 - s$ ). This way, the incentive to race can be traced to the trade-off between the two. Finally, the model enables us to quantify the specific cost-thresholds where different combinations of strategies become in the self-interest of states. Whereas Katzke and Futerman [22] and Young [39] characterize the race as a trust dilemma, and Abraham et al. [1] compare this with the prisoner’s dilemma paradigm, we show that the race can occupy two more strategic worlds than the the trust dilemma or the prisoner’s dilemma: the Safe Harmony world and the Subversion world.

We operationalize our model as a competition between two states.<sup>56</sup> Each state either chooses to race towards ASI or to pause conditionally, which result in one of three outcomes. First, a *safe win*: the player maintains safety and achieves supremacy. Secondly, a *safe loss*: the rival wins, but the player remains safe. Here, the player does not win geopolitical supremacy, but they might still enjoy spill-over effects. Finally, *catastrophe* might occur: one or both sides rush, safety fails, and the outcome is ruinous for all.

To represent the strategic playing field in which a state chooses to race by continuing frontier development with current safety levels or to pause conditionally by halting frontier development until safety conditions are met, then resuming with maximal safety, we define the following variables: the capability lead, the size of the winner’s advantage, the cost of uncontrolled ASI and contest noise. These variables reflect both theoretical necessity and the realities of the current AI landscape.

1. *Capability Lead* ( $\Delta$ ). This variable represents the relative advantage

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<sup>5</sup>While the global stage is multipolar, assuming only two players is legitimized by the immense concentration of computational infrastructure and talent required for ASI.

<sup>6</sup>The formal model can be found in the Appendix.

one player holds in developing ASI by aggregating factors such as computational power, research progress, and GDP [2]. The gap quantifies the advantage of the Frontrunner relative to the Laggard. This determines the probability of winning, as a larger lead makes victory more likely if both players adopt the same strategy. We range this from 0 (perfect parity) to potentially infinity (perfect imparity).

*Justification:* This variable is essential because relative advantage determines the probability of success; a model without  $\Delta$  would miss the crucial asymmetry between a Frontrunner, who may pause from a position of strength, and a Laggard, who faces the temptation to cut corners to close the gap.

2. *Winner’s Advantage ( $W$ )*. This variable represents the value of the first-mover advantage. It captures, among other factors, hostility, spillover effects from developing ASI and the speed with which winning translates into a military advantage [7, 2]. The Winner’s Advantage ranges from 0 (no winner’s advantage) to 1 (winner-takes-all).

*Justification:* Including  $W$  allows the model to distinguish between a cooperative race and an adversarial one where the loser faces geopolitical subordination.

3. *Cost of Uncontrolled ASI ( $C$ )*. We define  $C$  as the disutility arising from the deployment of uncontrolled systems, measured relative to the benefit of a safe win (normalized to 1). Unlike traditional threats like nuclear weapons, unaligned ASI is a *public cost* because a loss of control threatens the developer and rival alike. We allow for  $C > 1$ , reflecting the possibility that, for example, civilizational collapse or human extinction represents a loss far exceeding any strategic gain [35, 4, 38, 26].

*Justification:* Defining  $C$  as a public cost that can exceed the value of a safe win is necessary to model the potentially catastrophic nature of the control problem, which differs from ordinary costs in the combination of global scale and irreversibility. Without it, the model would treat the uncontrolled ASI as a cost maximally as bad as being subordinated to the technological supremacy of the other player, rather than a risk that can override the benefit of a risky win.

4. *Contest Noise ( $\sigma$ )*. This variable captures how decisively a capability lead turns into a win. When  $\sigma$  is small, even a small lead makes vic-

tory almost certain; when  $\sigma$  is large, the race is close to a coin flip, regardless of who is ahead. Formally,  $\sigma$  scales the capability difference in a standard logistic contest success function [19, 21].

*Justification:* Progress towards ASI depends on research breakthroughs, engineering bottlenecks and chance, so a capability lead does not guarantee winning. Without  $\sigma$ , the model would treat any lead as decisive.

The potentially catastrophic public cost can be mitigated by focusing on *safety* next to capability [2]. A state that *races* continues frontier development with the current level of safety. Racing is described by two parameters. The *Safety Level* ( $s$ ) is the probability that the ASI of a racing winner remains under control. In our central scenario we set  $s = 0.85$ , i.e. there is a 15% risk of catastrophe if a racer wins.<sup>7</sup> The *Race Boost* ( $b$ ) is the capability advantage that racing gives over a pausing rival, measured in the same units as  $\Delta$ ; we set  $b = 0.15$  in our central scenario. A state that *pauses conditionally* halts frontier development until safety conditions are met and then resumes with maximal safety. A pausing winner therefore avoids catastrophe, but the delay concedes the boost  $b$  to a racing rival, which lowers the pausing state’s chance of winning. Safety thus acts as a tax on capability: a racing state trades a catastrophe risk of  $1 - s$  for a speed advantage  $b$ , and the ratio  $b/(1 - s)$  is the rate at which safety is exchanged for speed.

Together, these variables constitute a minimal set needed to analyze the pause question. With the playing field defined, we can now examine under which parameter values it is rational for a player to pause and when a pause is stable. In other words, we can examine under what conditions, for both states, the pause strategy optimizes their self-interest.

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<sup>7</sup>The value  $s = 0.85$  is an illustrative central scenario, not a structural estimate. Expert surveys estimate probabilities of extinction or of bad outcomes from advanced AI in general, they do not estimate the model-specific probability of catastrophe given that a racing state wins. We use them only to bracket plausible scenarios. Median forecasts for human extinction hover around 5%, while between 38% and 51.4% of more than 2,700 surveyed researchers assign at least a 10% probability to outcomes such as the permanent loss of human control [13, 17]. An earlier structured survey reported a cumulative 31% probability of bad or extremely bad outcomes [25]. These estimates face significant model uncertainty [14]. In section 3 we therefore report sensitivity results for  $s = 0.95$  and  $0.85$ .

### 3 Four Possible Strategic Worlds

To determine whether a policy is rational, a state must evaluate its best response to a rival’s policy. In our model, this decision hinges on a catastrophic threshold: the point where the potential cost of an unaligned ASI ( $C$ ) outweighs the expected payoff of the outcome of the race. By mapping these thresholds against the capability lead ( $\Delta$ ), we identify four distinct strategic worlds. These worlds are identified based on their combinations of stable outcomes and their size depends on the winner’s advantage ( $W$ ), contest noise ( $\sigma$ ), and the racing parameters  $s$  and  $b$ .<sup>8</sup>

- World 1 (Safe Harmony): In this world, it is rational for both players to pause. This occurs when the perceived cost of catastrophe is so high, that racing offers no meaningful marginal gain. For the Frontrunner, safety is nearly costless, for the Laggard, racing is extremely costly.
- World 2 (Preemption): In this world, the unique stable outcome is a race. Even if both states acknowledge that a mutual race will likely end in ruin, the cost of catastrophe is valued lower than the fear of the rival’s first-mover advantage. This world is largest at near-parity. When states are neck-and-neck the temptation to sprint via corner-cutting is at its peak.[2, 1, 8].<sup>9</sup>

While World 1 (Safe Harmony) and World 2 (Preemption) align with the findings in Armstrong et al. [2], our model reveals two additional strategic worlds. By shifting the set of strategies from a continuous set of safety-levels, to a more realistic discrete set of policy choices (safety vs race) and allowing catastrophic ruin to outweigh the value of a safe win ( $C > 1$ ), we derive the worlds of Trust and Subversion.

- World 3 (Trust): Here, coordination is the barrier. Both states prefer a mutual pause over a mutual race, but they fear being taken advantage of by pausing while the rival races to a risky victory. This is a classic coordination game where the outcome is determined by diplomatic trust and the signaling of intentions [23, 1, 18, 22].<sup>10</sup> Both mutual

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<sup>8</sup>Proposition 1 in the Appendix defines the four worlds formally.

<sup>9</sup>Note that the Prisoner’s Dilemma is included in this world, but is not identical to it.

<sup>10</sup>Many coordinating mechanisms have recently been proposed to pull both states to the mutual pause [16, 18, 31, 15].

pausing and mutual racing are equilibria, but the model itself does not determine on which one states will coordinate.

- **World 4 (Subversion):** This is an asymmetric world. The Frontrunner, fearing catastrophe and confronted with a high probability of winning, chooses to pause. However, the Laggard, seeing a small window to overtake a responsible Frontrunner and a sufficiently low cost of loss of control, chooses to race. Here, pausing mainly influences strategy through the race boost it concedes to the rival, so that caution acts as a handicap.

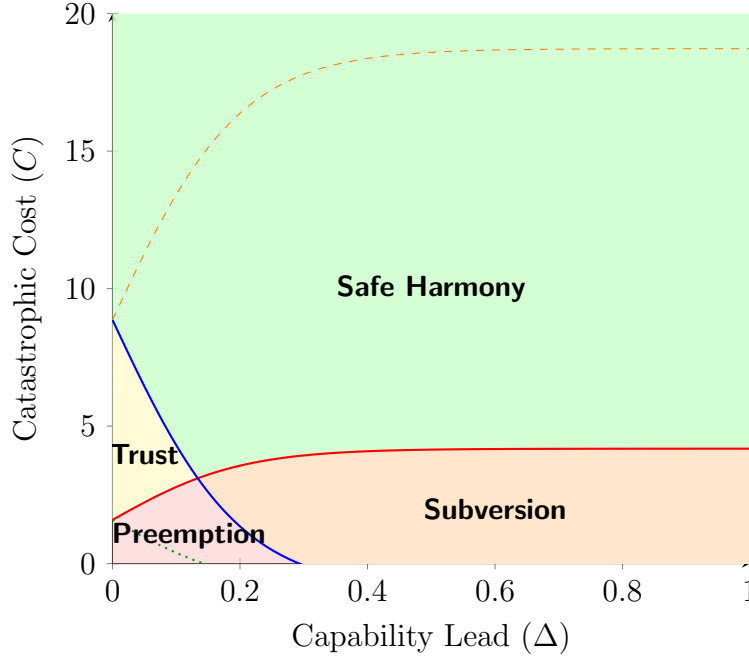


Figure 1: Decisive contest, winner takes all ( $W = 1$ ,  $\sigma = 0.1$ ,  $s = 0.85$ ,  $b = 0.15$ )

Figures 1–5 map the four strategic worlds as an interaction between the capability lead and catastrophic risk together with four threshold functions, two of which ( $LC$  and  $FUB$ ) delineate the worlds. The Cooperation Thresholds (Green/Red, for Frontrunner and Laggard respectively) determine the cost of loss of control at which the Frontrunner or Laggard switches from racing to pausing, *provided the rival is also pausing*. In contrast, the Unilateral

Break Thresholds (Blue/Orange, for Frontrunner and Laggard respectively) identify the point at which a player chooses to pause *if the rival continues to race*.

Strategically, any coordinate located above a player’s threshold indicates that the catastrophic risk is high enough to make a pause their rational best response. The Safe Harmony (Green) world exists where both players are located above their respective thresholds and both pause. In the Trust world (Yellow), the players’ incentives overlap such that mutual pausing is preferred but requires coordination to avoid the sucker’s payoff. The world of Subversion (Orange) represents an asymmetric state where the Frontrunner finds it rational to pause while the Laggard still perceives a higher utility in racing to close the gap. Finally, Preemption (Red) occurs when both players fall below their thresholds, meaning the fear of being overtaken or ‘suckered’ outweighs the fear of catastrophe, forcing a mutual and potentially ruinous race.

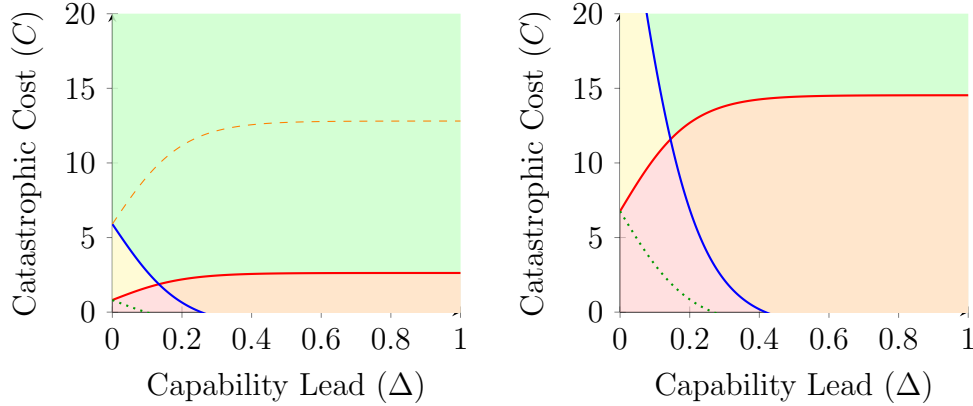


Figure 2: Partial winner’s advantage Figure 3: Safer racing ( $W = 1$ ,  $\sigma = 0.15$ ) ( $W = 0.7$ ,  $\sigma = 0.1$ ,  $s = 0.85$ ,  $b = 0.1$ ,  $s = 0.95$ ,  $b = 0.15$ )

The sizes of the possible worlds depend on the winner’s advantage, the safety level, the contest noise, and the racing boost.<sup>11</sup> In line with [2, 8], we find that a decrease in the size of the winner’s advantage makes a mutual pause rational for a wider range of costs (compare Figure 1 and 2). Secondly, a higher safety level for racers makes a pause harder to sustain (Figure 1 and

<sup>11</sup>For the proves of these effects, see Proposition ?? in the Appendix.

3). If racing carries a 5% ( $s = 0.95$ ) rather than a 15% risk of catastrophe, with the same speed advantage, the cost at which a mutual pause becomes the unique stable outcome at parity rises, and Subversion extends to much higher costs at larger leads. This is very intuitive, since low risk makes a pause superfluous with respect to the parameters in the model. Note that  $W$  only rescales the thresholds and leaves the shape of the worlds unchanged, while  $s$  does so for the cooperative thresholds, but additionally scales the unilateral break thresholds by  $s$ .

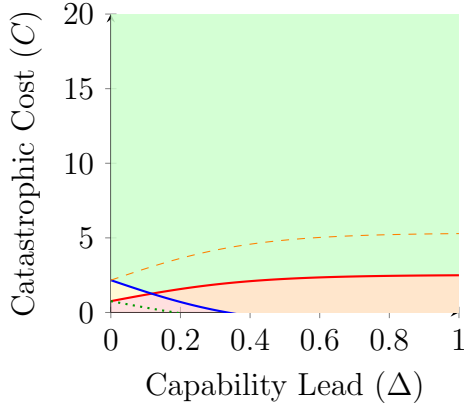


Figure 4: Noisier contest ( $W = 1$ ,  $\sigma = 0.2$ ,  $s = 0.85$ ,  $b = 0.15$ )

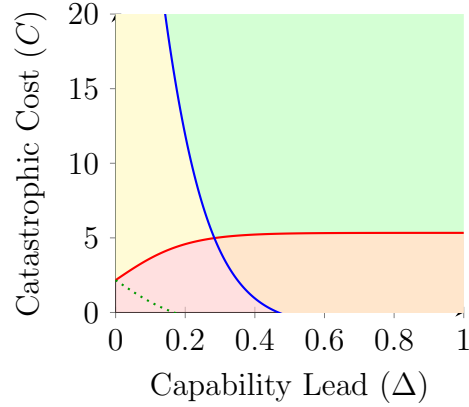


Figure 5: Larger race boost ( $W = 1$ ,  $\sigma = 0.1$ ,  $s = 0.85$ ,  $b = 0.30$ )

Thirdly, decisiveness incentivizes racing. In Figure 1 ( $\sigma = 0.1$ ), a small lead makes victory nearly certain, so the Laggard's incentive to race in order to overtake is strong (Figure 1 and 4). In Figure 4 ( $\sigma = 0.2$ ), a noisier contest smooths the thresholds and the Safe Harmony region grows because racing is less likely to translate into a win. Fourthly, a larger race boost also makes a pause harder to sustain, above all near parity (Figure 1 and 5). With  $b = 0.30$ , a pausing state is almost certain to be overtaken, so the Trust world expands and the cost threshold to enter the Safe Harmony World becomes very high. The race boost changes the shape of the worlds, just as  $\sigma$  does, because  $b$  enters the thresholds only through the ratio  $b/\sigma$ .

Finally, since safety acts as a tax,  $s$  and  $b$  need not vary independently. If it is taxed one-for-one ( $b = 1 - s$ ), a riskier race is also a faster one and near parity the effect is strongest. The cost at which a mutual pause becomes the unique stable outcome rises when safety decreases, which is the opposite of

the direction found when  $b$  is held fixed. In all cases the four worlds keep the same arrangement, so the qualitative conclusion is robust. However, the cost of loss of control required for a pause is sensitive to all four parameters.

## 4 Rising Concern about Loss of Control

In the previous section, we demonstrated that a pause-strategy optimizes a state’s self-interest when the cost of loss of control ( $C$ ) rises sufficiently high, relative to the other parameters. However, states act not on the true cost of loss of control, but on their perception of it. Hence the question: What about the *perception* of the cost of loss of control? In what direction is it evolving? The indicators below are descriptive. They show that concern about loss of control has become more salient among experts, the public and governments.<sup>12</sup>

A series of expert statements, governmental actions, and public opinion surveys are consistent with a rise in the perceived cost of uncontrolled AI, which in our model would enlarge the strategic space for cooperation. In March 2023, the Future of Life Institute published an open letter calling for a six-month pause on the training of AI systems more powerful than GPT-4. The letter attracted over 30 000 signatories, including Yoshua Bengio, Stuart Russell, Elon Musk, and Steve Wozniak [9]. Notably, the letter explicitly warned of race-to-the-bottom dynamics in which labs are incentivized to cut safety corners to deploy products more quickly, which corresponds to the Preemption world in our model. Just two months later, the Center for AI Safety released a one-sentence statement declaring that mitigating the risk of extinction from AI should be treated as a global priority alongside pandemics and nuclear war. This statement was signed by hundreds of leading researchers, including Turing Award laureates Geoffrey Hinton and Yoshua Bengio, as well as the CEOs of OpenAI, Anthropic, and DeepMind [5].

Public opinion data suggest that these concerns extend well beyond the research community. A YouGov poll from 2023 found that 43% of respondents were at least somewhat concerned about AI causing humanity’s extinction

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<sup>12</sup>Note that they do not measure  $C$ , which in our model is a state’s valuation of catastrophe relative to a safe win. Because they come from different populations, question wordings and dates, they cannot be read as a time series for a single variable, and they do not show that any pair of states is close to one of the thresholds in Section 3. We present them as evidence consistent with a rise in perceived  $C$ , not as a test of the model.

[27]. Furthermore, about 60% supported the FLI’s call for a six month pause on AI development [9]. A Reuters/Ipsos poll corroborated these findings, as 61% of Americans believed that AI poses risks to humanity [36].

Increased concern among researchers and the public may shift the discourse, but it does not necessarily translate into a corresponding shift in policy-making. However, in the case of advanced AI, the expert warnings also triggered rapid political action. In October 2023, the Biden administration signed the first major executive order specifically targeting AI safety, requiring developers to share safety test results with the federal government. The following month, the UK government hosted the first-ever global AI safety summit at Bletchley Park. Twenty-eight countries signed the Bletchley Declaration, which raised concerns about catastrophic risks from frontier AI systems [37]. Furthermore, the summit catalyzed the creation of dedicated safety institutions. The UK established its AI Safety Institute in November 2023, with the US following through a parallel body housed within NIST [30]. By the Seoul AI Summit in May 2024, ten countries plus the European Union had agreed to form an international network of AI Safety Institutes, and South Korea announced its own AI Safety Research Center [6].

This progression, from expert alarm in 2023 to institutional build-out by 2024, suggests that concern about loss of control has moved from the fringe of the AI safety community to a mainstream political and institutional priority. It does not show by how much states’ perceived cost of loss of control has changed, nor where states currently stand relative to the thresholds identified in Section 3. What it does suggest is that the parameter on which our results turn is not fixed: if the perceived cost of loss of control continues to rise while the other variables remain stable, the conditions under which a pause is in a state’s self-interest could come within reach. In future work, a more direct measure of perceived  $C$ , for example through repeated surveys of policymakers with consistent questions based on this model, could provide an accurate estimation of  $C$ .

## 5 Limitations and Extensions

In game theory one must make trade-offs between simplicity and realism. We need to carefully select which strategic factors to consider, while also being able to understand their impact. Inevitably, then, one leaves important information out. In future work, the following assumptions could be altered,

which may affect the game:

- *Dynamic game.* Our model assumes the competition towards ASI is played in one shot. However, it is more realistic to assume the game is played dynamically, such that the game consists of a sequence of moves until a player develops ASI, where the development of ASI becomes more likely through time. This temporal dimension might influence the rationality of the pause strategy. For example, we can expect pausing to be rational until the marginal gains from additional safety become lower than the marginal gains from preempting the opponent [24].
- *n-player game.* Our model assumes the game to consist of two players. Although in current circumstances this is a legitimate simplification, more states might build significant capacity to develop ASI in the future. It is thus interesting to extend our two-player game to an  $n$ -player game in future work. From Armstrong et al.’s paper [2], we can expect the presence of more players to have a negative effect on the size of the worlds where a pause strategy is rational.
- *n-ary choice.* In our model we assumed a binary policy choice to be made. Although policy choice is coarse grained, hence a discrete set of strategies is more realistic than a continuous one, it would be more realistic to add a third strategy to each player’s strategy set. This extra option could be an increase of a state’s focus on safety, while continuing the development of ASI. We might expect a ‘Goldilocks-zone’ to emerge where, under very specific circumstances, this strategy is rational.
- *Duration of the pause.* We model the conditional pause in reduced form, that is, the race boost  $b$  summarizes how much ground a racing rival gains while the pausing state waits for safety conditions to be met. A more fine-grained model would make the pause explicit, with a duration  $\tau$  and a capability gain per unit of time  $\kappa$ , so that a racing rival’s head start is  $\kappa\tau$ , and with a residual risk that declines with the safety work done during the pause. Such a model would make it possible to study how the stability of a pause depends on its length and what the ideal length is to resume development.
- *Beliefs.* In our model we assume the cost of unaligned ASI to be an exogenous variable, valued identically by both players. However, in future work, it might be interesting to introduce each player’s, potentially

asymmetric belief of the cost. We conjecture that different beliefs about the cost of unaligned ASI and the accessibility of information about the other’s beliefs will have effects on the rationality of the pause strategy.

- *Nested games.* We treat nations themselves as rational actors. In reality, policy decisions arise through the actions of a large number of agents within each nation. And while there are nation-level games being played, there are also countless nested games among the actors within the nations. For instance, the US has a two-party system with a four-year presidential election cycle. The ruling party is thus playing a strategic game to retain executive power. This lends importance to short-term policy decisions, which may contradict nation-level long-term rational decisions.
- *Multiple Development.* In our model, we assume the race stops when one of the players develops ASI. This grounds the assumption that the safety-level of the winning player determines the probability of loss of control. However, in reality a player who lost the game might still try to develop their own ASI, which would have an impact both on the strategic choice of each player and the probability of loss of control. In future work, our model can be extended to account for a race that can continue after one player develops ASI.

These and other complications can make for a deeper understanding of the dynamics of ASI-related rational choice, though their addition should be weighed against possible increases in model opacity and decreases in broad applicability, both of which can be side effects of increased complexity.

## 6 Conclusion

This paper has demonstrated that a pause on the development of ASI is not necessarily contrary to a state’s self-interest, as critics often claim. Through game-theoretic modeling, we identified four possible strategic worlds: Safe Harmony, Trust, Subversion, and Preemption. The crucial variable is the estimated cost of losing control over an unaligned ASI. When states assess that the potential catastrophic consequences, ranging from permanent loss of human autonomy to extinction, exceed the geopolitical benefits of technological

supremacy, relative to technological uncertainty and the safety level, strategic space emerges for a pause. Our analysis shows that a smaller size of the winner’s advantage between states and a degree of uncertainty about which state wins the race increases the self-interested basis for cooperation. Finally, based on empirical indicators since 2023, we suggest that although the current geopolitical reality remains dominated by competitive dynamics, the conditions under which a conditional pause becomes in a state’s self-interest may be closer than commonly assumed.

## Author Contributions

Edward Roussel: Conceptualization, Methodology, Formal Analysis, Writing - Original Draft. Lode Lauwaert: Conceptualization, Methodology, Writing - Original Draft. Torben Swoboda: Conceptualization, Writing - Original Draft. Grant Ramsey: Conceptualization, Writing - Original Draft. Risto Uuk: Conceptualization, Writing - Reviewing and Editing, Leonard Dung: Conceptualization, Writing - Reviewing and Editing. Anthony Aguirre: Conceptualization, Writing - Reviewing and Editing.

## Acknowledgements

We would like to thank Prof. Dr. Sylvia Wenmackers and Prof. Dr. Luc Lauwers for their feedback on the mathematical model grounding this paper. We also thank the members of the Chair Ethics and AI at KU Leuven, members of the Future of Life Institute and an anonymous reviewer for their feedback and discussion on this paper. This research was funded by the Research Foundation Flanders (FWO), Grant No. 1101426N.

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## Appendix

### The Model

We formalize the strategic model described in Section 2 as a two-player game with players  $i \in \{1, 2\}$ , the Frontrunner and the Laggard. Let  $\Delta \geq 0$  denote the Frontrunner’s relative capability advantage. Each player chooses a safety policy  $s_i$ , where  $s_i = 1$  represents a conditional pause (halting frontier development until it can resume with maximal safety), and  $s_i = s \in (0, 1)$  represents a race.

We model the probability of a win as a logistic contest function [19, 21], where  $\sigma > 0$  is contest noise. The probability that the Frontrunner wins when both players choose the same strategy is

$$P_{win}(\Delta) = P = \frac{1}{1 + e^{-\Delta/\sigma}},$$

and the probability that the Frontrunner loses (and the Laggard wins) is

$$P_{lose}(\Delta) = 1 - P = \frac{1}{1 + e^{\Delta/\sigma}}.$$

As  $\sigma \rightarrow 0$ , the outcome becomes deterministic, so that the smallest capability advantage results in a sure win and as  $\sigma \rightarrow \infty$ , a capability lead becomes less decisive and the winning probability approaches a coin toss.

A player who races gains a speed advantage  $b > 0$  over a pausing rival, but accepts the lower safety level  $s$ . We define the race Boost Constant as

$$B = \frac{b}{\sigma}.$$

The probability of winning when racing against a pausing opponent ( $P^R$ ) and when pausing against a racing opponent ( $P^S$ ) are

$$P^R = \frac{P}{P + (1 - P)e^{-B}}, \quad P^S = \frac{P}{P + (1 - P)e^B},$$

which equal  $1/(1 + e^{-(\Delta+b)/\sigma})$  and  $1/(1 + e^{-(\Delta-b)/\sigma})$ .

A player gets a payoff 1 if they win the race safely and  $1 - W$  if they lose safely, where  $W \in [0, 1]$  is the size of the winner's advantage. We define the cost of uncontrolled ASI as  $C$ . Since the utility of winning is normalized to 1,  $C$  is interpreted relative to this benefit. Contrary to Armstrong et al. [2], we allow  $C > 1$ . The winner's safety policy determines whether control is kept with the winner's probability  $s_i$ . The expected utility of player  $i$  is

$$EU_i(s_i, s_j) = P(i \text{ wins}) [s_i - (1 - s_i)C] + P(i \text{ loses}) [s_j(1 - W) - (1 - s_j)C].$$

## Derivation of Strategic Thresholds

A player pauses if the expected utility of pausing ( $s_i = 1$ ) is at least that of racing ( $s_i = s$ ). The thresholds below use four identities, which follow directly from the definitions of  $P^R$  and  $P^S$ :

$$\begin{aligned} 1 - \frac{P}{P^R} &= (1 - P)(1 - e^{-B}), & \frac{P}{P^S} - 1 &= (1 - P)(e^B - 1), \\ \frac{P - P^S}{1 - P^S} &= P(1 - e^{-B}), & \frac{P^R - P}{1 - P^R} &= P(e^B - 1). \end{aligned}$$

**Proposition 1 (Strategic thresholds).** *Let  $P = P_{win}(\Delta)$  and  $1 - P = P_{lose}(\Delta)$ , and resolve ties in favor of pausing.*

(i) *Frontrunner Cooperation (FC). The Frontrunner's best response to a pausing Laggard is to pause iff*

$$C \geq \frac{W}{1 - s} (1 - e^{-B}) P_{lose}(\Delta) - 1. \quad (1)$$

(ii) *Frontrunner Unilateral Break (FUB). The Frontrunner's best response to a racing Laggard is to pause iff*

$$C \geq \frac{sW}{1 - s} (e^B - 1) P_{lose}(\Delta) - 1. \quad (2)$$

(iii) *Laggard Cooperation (LC). The Laggard's best response to a pausing Frontrunner is to pause iff*

$$C \geq \frac{W}{1 - s} (1 - e^{-B}) P_{win}(\Delta) - 1. \quad (3)$$

(iv) *Laggard Unilateral Break (LUB). The Laggard's best response to a racing Frontrunner is to pause iff*

$$C \geq \frac{sW}{1-s} (e^B - 1) P_{win}(\Delta) - 1. \quad (4)$$

*Proof.* (i) If the Laggard pauses, the Frontrunner wins with probability  $P$  when it pauses and  $P^R$  when it races. It pauses iff  $EU_F(1, 1) \geq EU_F(s, 1)$ :

$$P + (1 - P)(1 - W) \geq P^R[s - (1 - s)C] + (1 - P^R)(1 - W).$$

Rearranging gives

$$P^R(1 - s)(1 + C) \geq W(P^R - P),$$

so

$$C \geq \frac{W}{1-s} \left(1 - \frac{P}{P^R}\right) - 1.$$

Substituting

$$1 - P/P^R = (1 - P)(1 - e^{-B})$$

yields (1).

(ii) If the Laggard races, the Frontrunner wins with probability  $P^S$  when it pauses and  $P$  when it races. It pauses iff  $EU_F(1, s) \geq EU_F(s, s)$ :

$$P^S + (1 - P^S)[s(1 - W) - (1 - s)C] \geq P[s - (1 - s)C] + (1 - P)[s(1 - W) - (1 - s)C].$$

Rearranging gives

$$P^S(1 - s)(1 + C) \geq sW(P - P^S),$$

so

$$C \geq \frac{sW}{1-s} \left(\frac{P}{P^S} - 1\right) - 1.$$

Substituting

$$P/P^S - 1 = (1 - P)(e^B - 1)$$

yields (2).

(iii) If the Frontrunner pauses, the Laggard wins with probability  $1 - P$  when it pauses and  $1 - P^S$  when it races. It pauses iff  $EU_L(1, 1) \geq EU_L(s, 1)$ :

$$(1 - P) + P(1 - W) \geq (1 - P^S)[s - (1 - s)C] + P^S(1 - W).$$

Rearranging gives

$$(1 - P^S)(1 - s)(1 + C) \geq W(P - P^S),$$

so

$$C \geq \frac{W}{1 - s} \cdot \frac{P - P^S}{1 - P^S} - 1.$$

Substituting

$$(P - P^S)/(1 - P^S) = P(1 - e^{-B})$$

yields (3).

(iv) If the Frontrunner races, the Laggard wins with probability  $1 - P^R$  when it pauses and  $1 - P$  when it races. It pauses iff  $EU_L(1, s) \geq EU_L(s, s)$ :

$$(1 - P^R) + P^R[s(1 - W) - (1 - s)C] \geq (1 - P)[s - (1 - s)C] + P[s(1 - W) - (1 - s)C].$$

Rearranging gives

$$(1 - P^R)(1 - s)(1 + C) \geq sW(P^R - P),$$

so

$$C \geq \frac{sW}{1 - s} \cdot \frac{P^R - P}{1 - P^R} - 1.$$

Substituting

$$(P^R - P)/(1 - P^R) = P(e^B - 1)$$

yields (4). □

Since  $P_{win}(\Delta) \geq P_{lose}(\Delta)$  for  $\Delta \geq 0$ , the Frontrunner's thresholds lie below the Laggard's. Since  $e^B - 1 = e^B(1 - e^{-B})$ , each unilateral break threshold lies above the corresponding cooperation threshold whenever  $se^B \geq 1$ , which holds for all parameter values used in this paper. A mutual pause is then stable iff  $C \geq LC$ , a mutual race iff  $C < FUB$ , and a pausing Frontrunner with a racing Laggard iff  $FUB \leq C < LC$ ; a pausing Laggard with a racing Frontrunner is never stable. This gives the four worlds of Section 3: Safe Harmony ( $C \geq \max\{LC, FUB\}$ ), Trust ( $LC \leq C < FUB$ ), Subversion ( $FUB \leq C < LC$ ) and Preemption ( $C < \min\{LC, FUB\}$ ).

**Proposition 2 (Sensitivity of the strategic worlds).** *Let  $W > 0$ ,  $\Delta \geq 0$  and  $se^B \geq 1$ , so that a mutual pause is the unique stable outcome iff  $C \geq \max\{LC, FUB\}$ , and Trust occurs iff  $LC \leq C < FUB$ .*

- (i) *Winner's advantage and safety level.  $LC$  and  $FUB$  increase in  $W$  and in  $s$  (holding  $b$  fixed). Moreover,  $1 + LC$  and  $1 + FUB$  are proportional to  $W$ , and to  $1/(1 - s)$  and  $s/(1 - s)$  respectively.*
- (ii) *Contest noise.  $\max\{LC, FUB\}$  decreases in  $\sigma$ . That means a noisier contest expands Safe Harmony at every capability gap.*
- (iii) *Race boost.  $\max\{LC, FUB\}$  increases in  $b$ . Trust occurs only at gaps  $\Delta < b + \sigma \ln s$ , and at each such gap the range of costs for which Trust occurs, from  $LC$  to  $FUB$ , widens as  $b$  increases.*
- (iv) *Safety tax. If  $b = 1 - s$ , the cost from which a mutual pause is the unique stable outcome at parity increases as  $s$  decreases iff*

$$\frac{Be^B}{e^B - 1} > \frac{1}{s}.$$

*Proof.* By (3) and (2),

$$LC = \frac{W}{1-s}(1 - e^{-B})P - 1, \quad FUB = \frac{sW}{1-s}(e^B - 1)(1 - P) - 1,$$

with  $P = P_{win}(\Delta)$  and  $B = b/\sigma$ .

(i)  $P$  and  $B$  do not depend on  $W$  or  $s$ , and  $W/(1 - s)$  and  $sW/(1 - s)$  increase in both  $W$  and  $s$ .

(ii)  $LC$  decreases in  $\sigma$ , because  $1 - e^{-b/\sigma}$  decreases in  $\sigma$  and  $P = 1/(1 + e^{-\Delta/\sigma})$  does not increase in  $\sigma$  for  $\Delta \geq 0$ . For  $FUB$ , write  $1 + FUB = \frac{sW}{1-s} g(1/\sigma)$  with  $g(u) = (e^{bu} - 1)/(1 + e^{\Delta u})$ . Then

$$\frac{d \ln g}{du} = \frac{b e^{bu}}{e^{bu} - 1} - \frac{\Delta e^{\Delta u}}{1 + e^{\Delta u}} > b - \Delta,$$

so for  $\Delta < b$  also  $FUB$  decreases in  $\sigma$ , and hence so does  $\max\{LC, FUB\}$ . By part (iii),  $LC > FUB$  iff  $\Delta > b + \sigma \ln s$ , which holds for every  $\sigma$  when  $\Delta \geq b$ , since  $\ln s < 0$ . In that case  $\max\{LC, FUB\} = LC$ , which decreases in  $\sigma$ .

(iii)  $1 - e^{-B}$  and  $e^B - 1$  increase in  $B$ , and  $P$  does not depend on  $b$ , so  $LC$  and  $FUB$  increase in  $b$ . Next,  $LC < FUB$  iff  $P/(1 - P) < se^B$ , i.e. iff  $e^{\Delta/\sigma} < se^{b/\sigma}$ , i.e. iff  $\Delta < b + \sigma \ln s$ . The width of the Trust range is

$$FUB - LC = \frac{W}{1 - s} [s(1 - P)(e^B - 1) - P(1 - e^{-B})],$$

whose derivative with respect to  $B$  is  $\frac{W}{1 - s} [s(1 - P)e^B - Pe^{-B}]$ . This is positive whenever Trust occurs, because then  $Pe^{-B} < P < s(1 - P)e^B$ .

(iv) At  $\Delta = 0$ ,  $P = 1/2$  and  $FUB \geq LC$  because  $se^B \geq 1$ , so the threshold is  $FUB$ . With  $x = 1 - s$  and  $B = x/\sigma$ ,  $1 + FUB = W(1 - x)(e^B - 1)/(2x)$ . Then

$$x \frac{d \ln(1 + FUB)}{dx} = \frac{Be^B}{e^B - 1} - \frac{1}{1 - x},$$

which is positive iff  $Be^B/(e^B - 1) > 1/s$ . A decrease in  $s$  is an increase in  $x$ .  $\square$